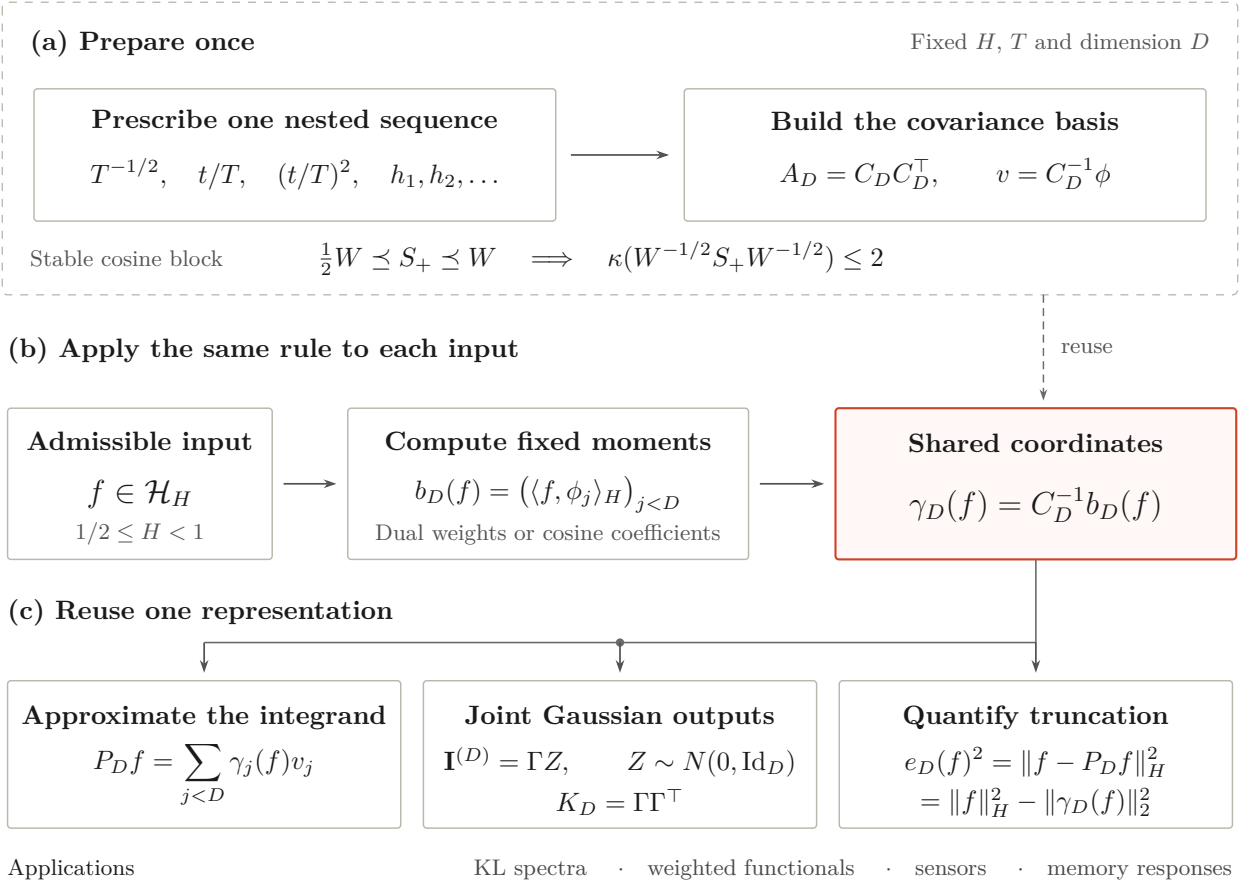

(a) Prepare once
Fixed H, T and dimension D
Prescribe one nested sequence
$T^{-1/2}, \quad t/T, \quad (t/T)^2, \quad h_1, h_2, \ldots$
Build the covariance basis
$A_D = C_D C_D^\top, \qquad v = C_D^{-1}\phi$
Stable cosine block
$\frac{1}{2}W \preceq S_+ \preceq W \implies \kappa(W^{-1/2} S_+ W^{-1/2}) \le 2$
reuse
(b) Apply the same rule to each input
Admissible input
$f \in \mathcal{H}_H$
$1/2 \le H < 1$
Compute fixed moments
$b_D(f) = \big(\langle f, \phi_j \rangle_H\big)_{j<D}$
Dual weights or cosine coefficients
Shared coordinates
$\gamma_D(f) = C_D^{-1} b_D(f)$
(c) Reuse one representation
Approximate the integrand
$P_D f = \sum_{j<D} \gamma_j(f) v_j$
Joint Gaussian outputs
$\mathbf{I}^{(D)} = \Gamma Z, \qquad Z \sim N(0, \mathrm{Id}_D)$
$K_D = \Gamma\Gamma^\top$
Quantify truncation
$e_D(f)^2 = \|f - P_D f\|_H^2$
$= \|f\|_H^2 - \|\gamma_D(f)\|_2^2$
Applications
KL spectra · weighted functionals · sensors · memory responses